

Note 4.7 Antidifferentiation--The Indefinite Integral

Antidifferentiation

- **Definition:**

A function F is an antiderivative of f on interval I if $F'(x) = f(x)$ for all x on I .

The process of finding antiderivatives is called antidifferentiation or indefinite integral.

- **Theorem:**

If F is an antiderivative of f on I , then the most general form of antiderivative of f on I is

$$F(x) + C$$

where C is an arbitrary constant.

And we use the symbol

$$\int f(x) dx$$

to represent the set of all antiderivatives of $f(x)$, and if $F'(x) = f(x)$, we write

$$\int f(x) dx = F(x) + C$$

$\int$ is the integral sign, f is the integrand, $\int f(x) dx$ is called the indefinite integral. (dx is the differentiation of x)

Corollary:

$$\frac{d}{dx} \left[\int f(x) dx \right] = f(x)$$

$$\int F'(x) dx = F(x) + C$$

Basic Integration Formulas

TABLE 4.2 Antiderivative formulas, k a nonzero constant

Function	General antiderivative	Function	General antiderivative
1. x^n	$\frac{1}{n+1}x^{n+1} + C, \quad n \neq -1$	8. e^{kx}	$\frac{1}{k}e^{kx} + C$
2. $\sin kx$	$-\frac{1}{k}\cos kx + C$	9. $\frac{1}{x}$	$\ln x + C, \quad x \neq 0$
3. $\cos kx$	$\frac{1}{k}\sin kx + C$	10. $\frac{1}{\sqrt{1-k^2x^2}}$	$\frac{1}{k}\sin^{-1} kx + C$
4. $\sec^2 kx$	$\frac{1}{k}\tan kx + C$	11. $\frac{1}{1+k^2x^2}$	$\frac{1}{k}\tan^{-1} kx + C$
5. $\csc^2 kx$	$-\frac{1}{k}\cot kx + C$	12. $\frac{1}{x\sqrt{k^2x^2-1}}$	$\sec^{-1} kx + C, \quad kx > 1$
6. $\sec kx \tan kx$	$\frac{1}{k}\sec kx + C$	13. a^{kx}	$\left(\frac{1}{k \ln a}\right)a^{kx} + C, \quad a > 0, a \neq 1$
7. $\csc kx \cot kx$	$-\frac{1}{k}\csc kx + C$		

- $$\int \sec x \, dx = \ln |\sec x + \tan x| + C$$

Algebraic Rules for Indefinite Integration

- The Constant multiple rule:

$$\int k f(x) \, dx = k \int f(x) \, dx$$

- The Sum Rule:

$$\int (f(x) \pm g(x)) \, dx = \int f(x) \, dx \pm \int g(x) \, dx$$